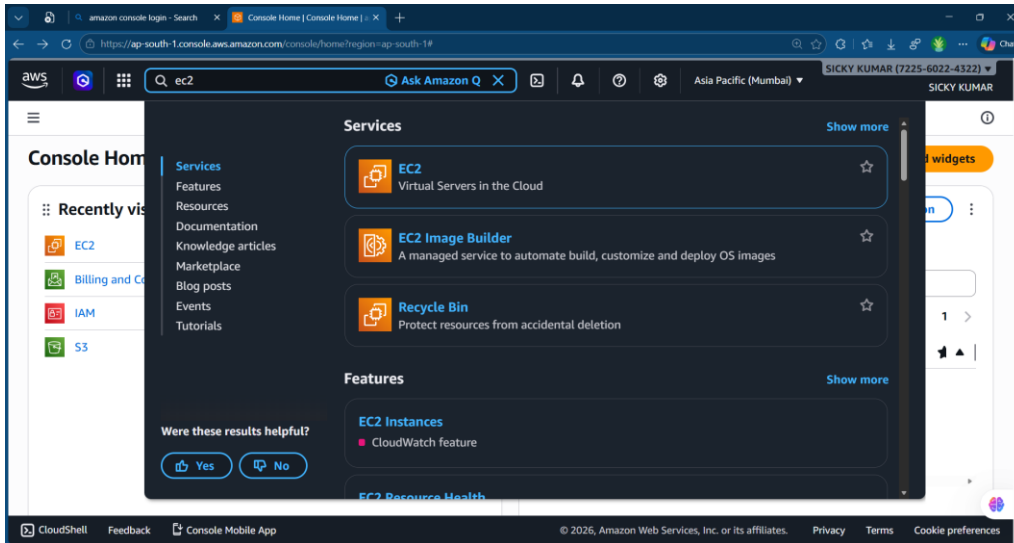


Assignment No: 10

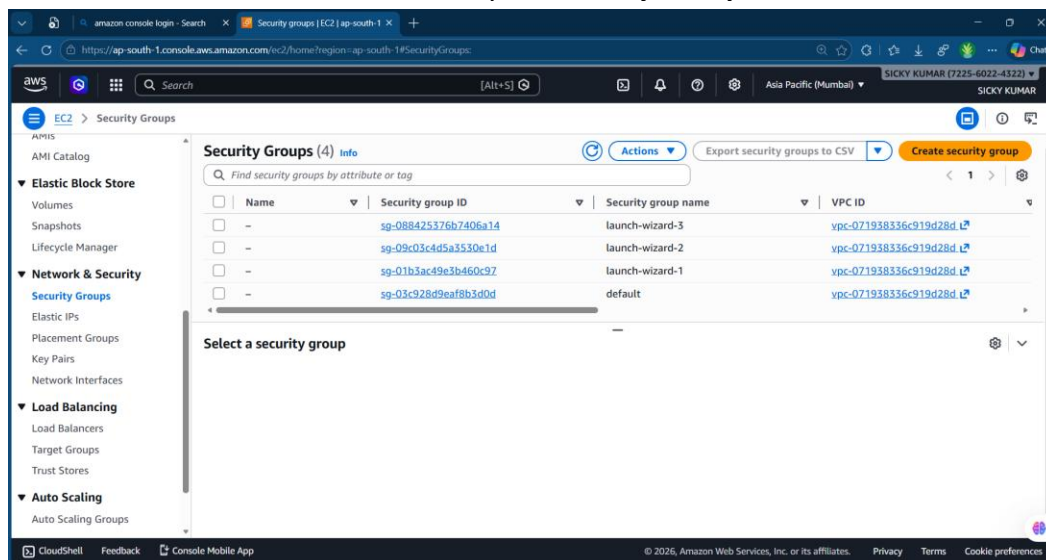
Problem Statement: Deploy a project from GitHub to EC2 by creating a new security group and user data.

Step 1: Open EC2 Service

From the AWS Management Console, search for EC2 and click on Elastic Compute Cloud (EC2).

**Step 2: Create a New Security Group**

EC2 Dashboard → Network & Security → **Security Groups** → Click **Create security group**

**Step 3: Configure Security Group Rules**

Security group name → Enter description →

Inbound rules → Add rules:

- SSH → Port 22 → Source 0.0.0.0/0
- HTTP → Port 80 → Source 0.0.0.0/0
- HTTPS → Port 443 → Source 0.0.0.0/0
- Custom TCP → Port 4000 → Source 0.0.0.0/0 →
Click Create security group (Security group created successfully)

Fill the Basic details:Security group name (sicky9304s)

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

sicky9304s

Name cannot be edited after creation.

Description [Info](#)

Allow SSH Access to Developer

VPC [Info](#)

vpc-071938336c919d28d

Set inbound rules → Add rules: SSH → Port 22 → Source 0.0.0.0/0

- HTTP → Port 80 → Source 0.0.0.0/0
- HTTPS → Port 443 → Source 0.0.0.0/0
- Custom TCP → Port 4000 → Source 0.0.0.0/0

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional	Actions
SSH	TCP	22	Any...		Delete
HTTPS	TCP	443	0.0.0.0/0		Delete
HTTP	TCP	80	0.0.0.0/0		Delete
Custom TCP	TCP	4000	0.0.0.0/0		Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Security group (sg-04df6fbc7635a1322 - sicky9304s) was created successfully

sg-04df6fbc7635a1322 - sicky9304s [Actions](#)

Details

Security group name sicky9304s	Security group ID sg-04df6fbc7635a1322	Description Allow SSH Access to Developer	VPC ID vpc-071938336c919d28d
Owner 722560224322	Inbound rules count 4 Permission entries	Outbound rules count 1 Permission entry	

[Inbound rules](#) [Outbound rules](#) [Sharing](#) [VPC associations](#) [Related resources - new](#) [Tags](#)

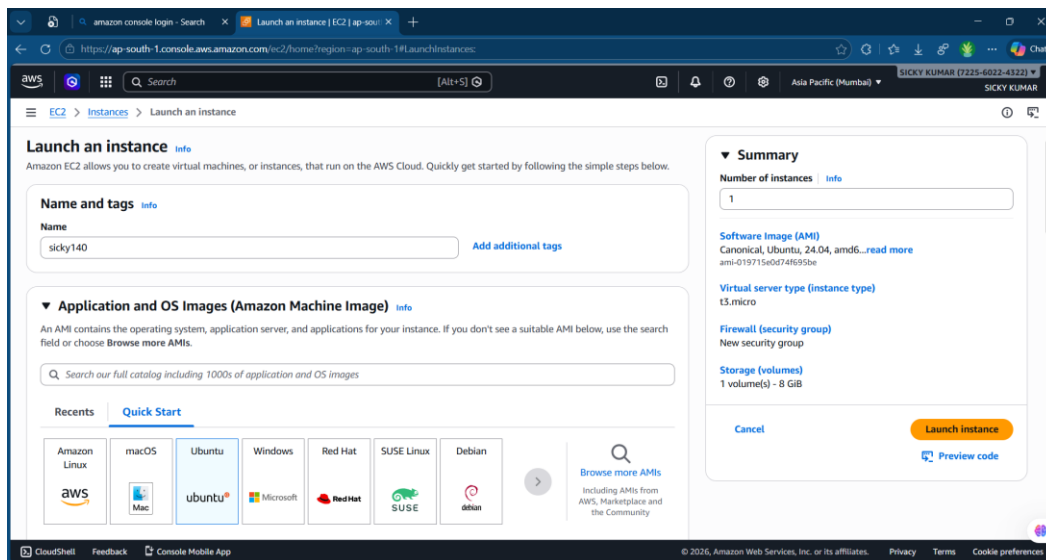
Inbound rules (4) [Manage tags](#) [Edit inbound rules](#)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0eab485c94abaa54	IPv4	Custom TCP	TCP	4000
-	sgr-001322f5511137c90	IPv4	SSH	TCP	22

Security group created successfully

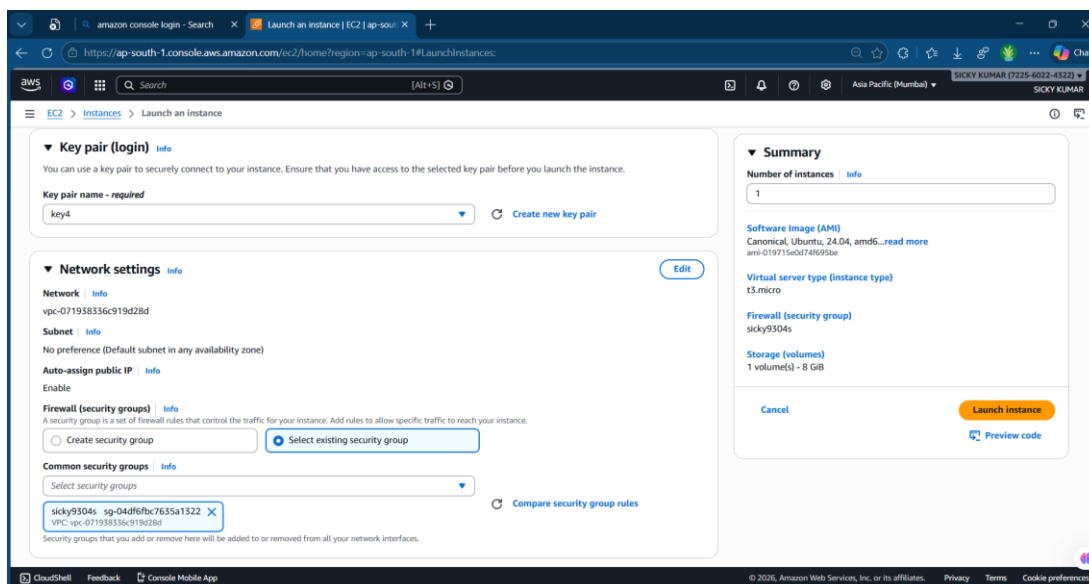
Step 3: Launch EC2 Instance

EC2 Dashboard → Instances → Launch instance → Enter instance name →
Application & OS Images → Select Ubuntu



Step 4: Configure Key Pair and Security Group

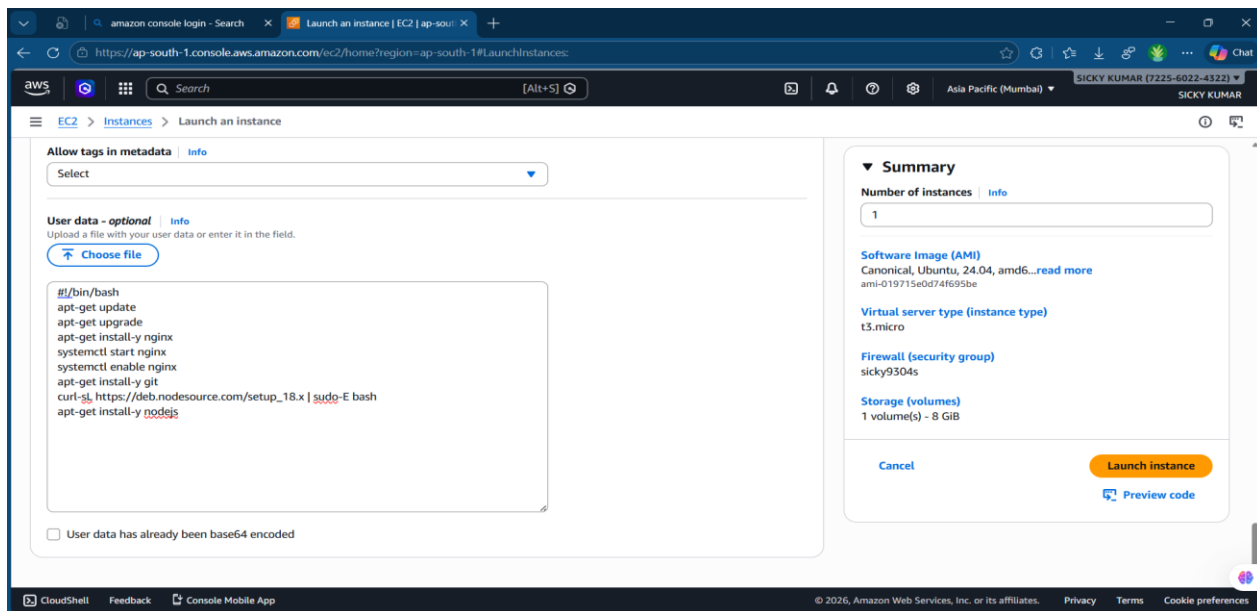
Key pair (login) → Select existing / create new key pair →
Network settings → Select **existing security group** (sicky9304s)



Step 5: Click on advance rule then Add User Data Script.

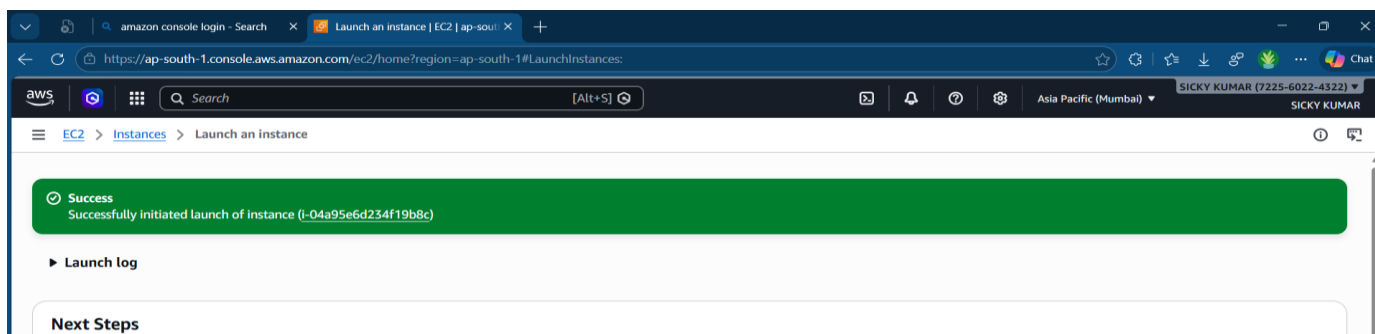
Advanced details → User data → write the following script:

```
apt-get update
apt-get upgrade -y
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash
apt-get install -y nodejs
```

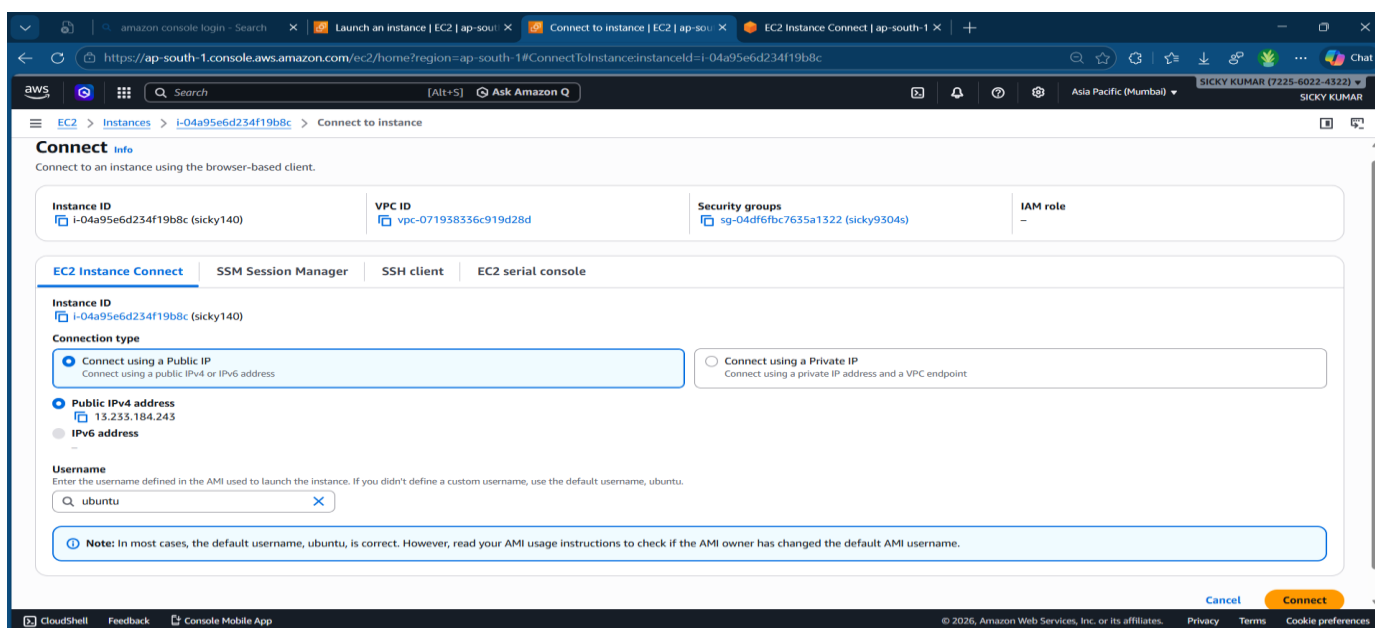


Step 6: Launch the Instance

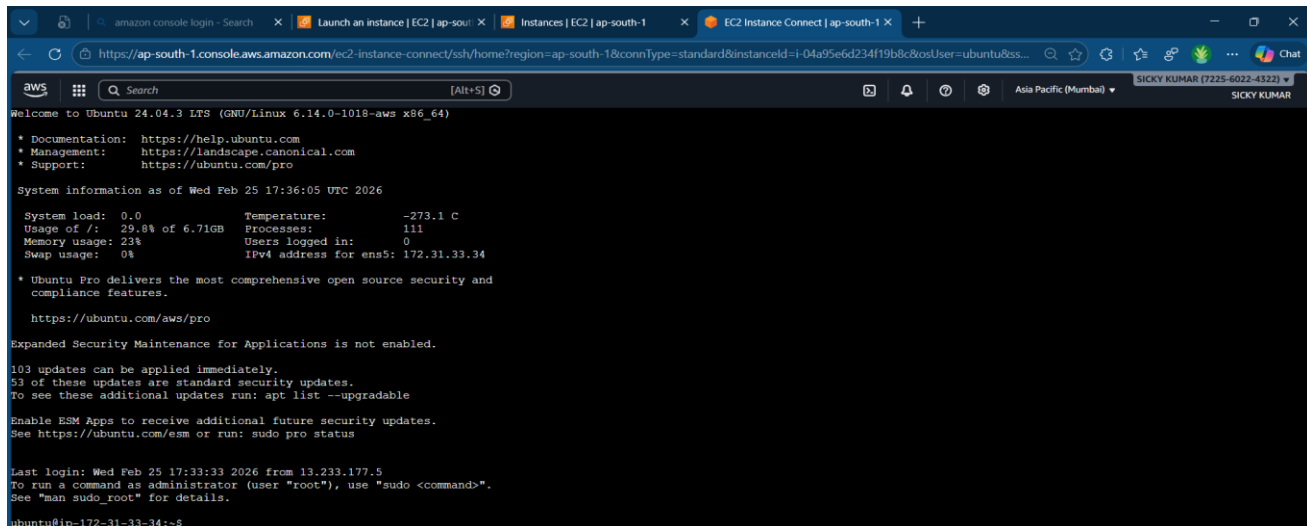
Click **Launch instance** → Instance created successfully → state shows **Running**



Step 7: After creating EC2 Instances click on "CONNECT"



Step 8: After connecting It show all the details of instance in new terminal.



```
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Wed Feb 25 17:36:05 UTC 2026

System load:  0.0      Temperature:   -273.1 C
Usage of /:   29.8% of 6.71GB      Processes:    111
Memory usage: 23%      Users logged in: 0
Swap usage:   0%        IPv4 address for ens5: 172.31.33.34

* Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.
  https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

103 updates can be applied immediately.
53 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Feb 25 17:33:33 2026 from 13.233.177.5
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-33-34:~$
```

Step 8: Verify Nginx Installation

EC2 → Instances → Select instance → Copy **Public IPv4 address(3.110.220.236)** →

Open new browser tab → Paste IP address →

Welcome to nginx page displayed (Nginx installed successfully)



Step 10: Clone Project from GitHub

Terminal → Run: git clone <https://github.com/Sicky9304/demo-app.git>

Step 11: Install Project Dependencies

Terminal → demo-app → npm install

```
Last login: Wed Feb 25 17:53:40 2026 from 13.233.177.3
ubuntu@ip-172-31-33-34:~$ git clone https://github.com/Sicky9304/demo-app.git
Cloning into 'demo-app'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (13/13), done.
remote: Total 13 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (13/13), 4.48 KiB | 1.49 MiB/s, done.
Resolving deltas: 100% (5/5), done.
ubuntu@ip-172-31-33-34:~$ ls
demo-app
ubuntu@ip-172-31-33-34:~$ cd demo-app
ubuntu@ip-172-31-33-34:~/demo-app$ ls
index.js  package.json
ubuntu@ip-172-31-33-34:~/demo-app$
```

Step 12: Start the Server

```
ubuntu@ip-172-31-33-34:~/demo-app$ npm install
added 224 packages, and audited 225 packages in 10s

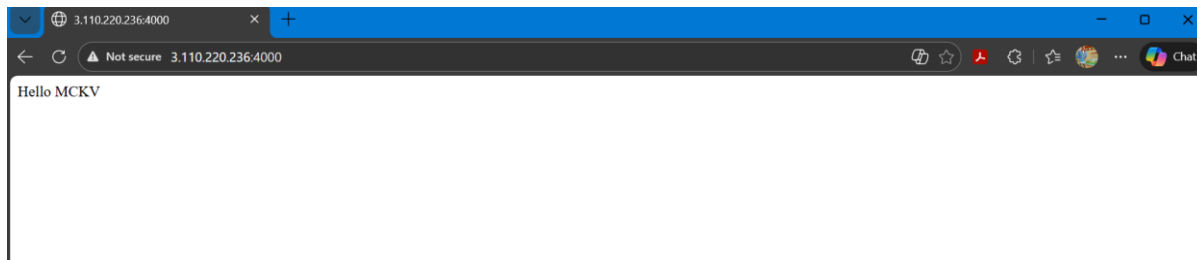
26 packages are looking for funding
  run `npm fund` for details

1 low severity vulnerability

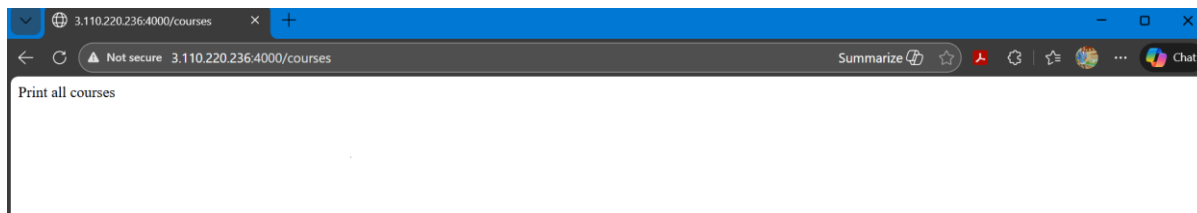
Some issues need review, and may require choosing
a different dependency.

Run `npm audit` for details.
ubuntu@ip-172-31-33-34:~/demo-app$ node index.js
Started server
```

Step 13: Now we go to the IP 3.110.220.236:4000 to check our hosted project



Step 14: Now we go to the IP 3.110.220.236:4000/courses to check our hosted project



[we have successfully hosted our project by using "NEW SECURITY GROUP"]